Ruiyi Fang

Homepage: https://raelynfang.github.io/ | GitHub | LinkedIn

ruiyifang@g.ecc.u-tokyo.ac.jp | Google Scholar

Education

- **The University of Tokyo** — Tokyo, Japan
  Doctor of Engineering in Mechanical Engineering — *Oct.2025 - Present*
  Advisor: Yuji Yamakawa
- **Central South University** — Changsha, China
  Master of Engineering in Artificial Intelligence — *Sep.2022 - Jun.2025*
  Advisor: Kai Wang
- **Chang'an University** — Xi'an, China
  Bachelor of Engineering in Automation — *Sep.2018 - Jun.2022*

Publications

- **Multi-step Difference-driven Domain Adversarial Network for Few-sample Fault Detection in Dynamic Industrial Systems**
  Ruiyi Fang, Kai Wang*, Xiaofeng Yuan, Zeyu Yang, Yalin Wang and Chunhua Yang
  Engineering Applications of Artificial Intelligence (EAAI), 2025
- **Unsupervised Domain Adversarial Network for Few-sample Fault Detection in Industrial Processes**
  Ruiyi Fang, Kai Wang*, Jing Li, Xiaofeng Yuan and Yalin Wang
  Advanced Engineering Informatics (AEI), 2024
- **Wasserstein Distance Based Domain Adversarial Autoencoder for Industrial Few-sample Fault Detection**
  Ruiyi Fang, Kai Wang*, Xiaofeng Yuan, Yalin Wang, and Chunhua Yang
  Asian Control Conference (ASCC), Jul. 2024 (Oral)

Research Experience

- **Few-Shot Imitation Learning for Robotic Manipulation** — *May.2025 - Present*
  - Researched on Imitation Learning (IL), Vision-Language-Action (VLA) models and Diffusion Policies to facilitate data-efficient, accurate and general robot manipulation. The proposed methods are validated through experiments on both simulation platforms and real-world robots under few-shot settings, demonstrating improved robustness and efficiency.
- **Transfer Learning based Fault Detection via Spatio-temporally Industrial Data** — *Sep.2022 - Apr.2025*
  - Researched on the theory and applications of Transfer Learning and Generative Adversarial Model with limited data. Incorporated self-attention mechanism to capture spatio-temporal features and designed a rebalanced training scheme, improving robustness and accuracy of fault detection in dynamic industrial processes.

Projects

- **VLA-based Robotic Manipulation for Real-world Daily Tasks** — *Nov.2025 - Jan.2026*
  - Established a VLA-based pipeline using the π_0 model and applied on LeRobot. Teleoperation was employed for data collection, while LCM and SSH were utilized to enable low-latency data transmission between remote server and local computer. The π_0 model was subsequently fine-tuned on the collected dataset, and the trained policy was deployed on the physical robot, achieving accurate manipulation performance for daily tasks.

Selected Awards & Honors

- **Outstanding Graduate** — *2025*
- **National Scholarship** — *2024*
- **Post-graduate First-Class Scholarship** — *2022,2023,2024*
- **Academic Excellence Scholarship** — *2018,2021*
- **"HUAWEI CUP" China Post-graduate Mathematical Contest In Modeling** Second Prize(Leader) — *2023*
- **Interdisciplinary Contest in Modeling (ICM)** Honorable Mention(Leader) — *2021*
- **National Mathematics Competition for College Students** Third Price — *2021*
- **National English Competition for College Students** Third Prize — *2021*
- **National English Translation Competition for College Students** First Prize(1%) — *2020*
- **Contemporary Undergraduate Mathematical Contest in Modeling** Second Prize(Leader) — *2020*

Skills

Programming Skills: Python/PyTorch, Matlab, C, C++, CAD, Java, LaTeX, HTML/CSS, Git
Robotic Simulation & Control Platforms: Isaac Sim, MuJoCo, ROS, Coppeliasim

Language

Mandarin (Native), **English** (Fluent), **Japanese** (Advanced)

Academic Services

Reviewer:
Knowledge-Based Systems (KBS), Asian Control Conference (ASCC)